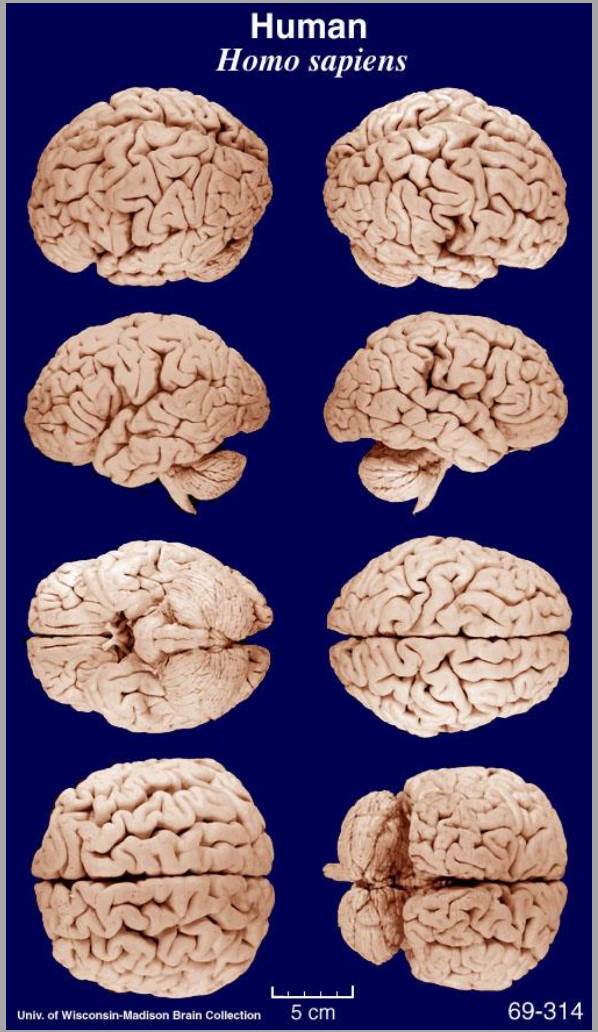
Session objectives

- Students will be able to describe the gross organization of the neocortex in relation to other structures in the brain as well as organization according to major lobes, fissures, sulci, and gyrii.
- Students will be able to describe the anatomy of neocortical connections including those associated with intracortical, intercortical, and subcortical circuits.
- Students will be able to describe the basic cytoarchitecture of the neocortex as it relates to its six layers.
- Students will be able to describe the anatomy of the major neuron types of the neocortex including the basic physiology of these neurons.
- Students will be able to describe the physiology of neocortical neurons as it relates to sensation, voluntary motor function, attention, and higher mental processes

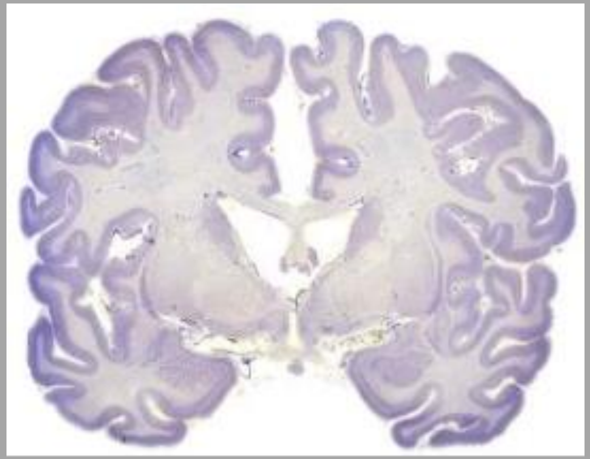
Neocortex: organized by sulci, gyri, & lobes.



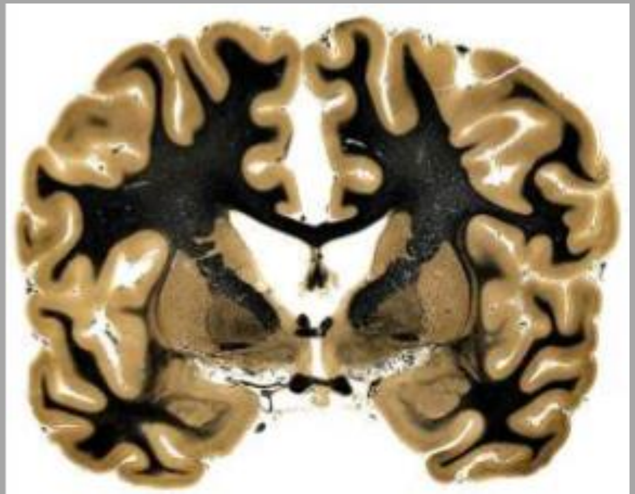
- Gray matter= cell bodies of neurons
- White matter = axons

Unstained

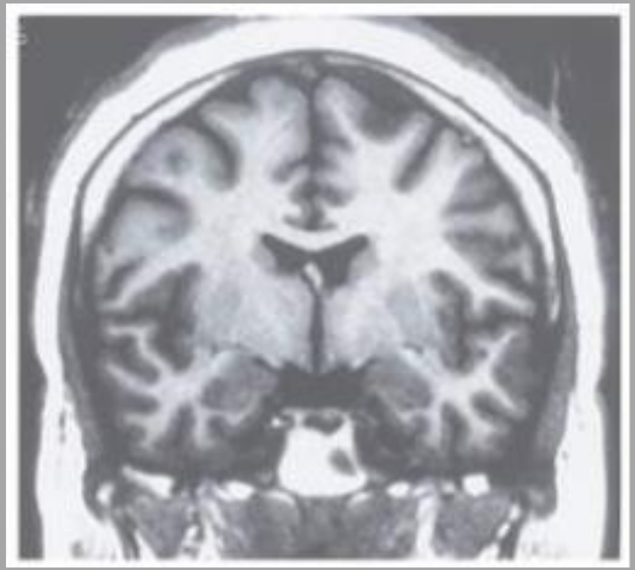
Cell body stain (Nissl) stains Nissl bodies, which are ER and ribosome of neurons



Axonal stain



MRI



In vivo

- It would take over a week to discuss every part of the neocortex. We are covering the major lobes and gyri, involved in sensation, perception behavior etc.

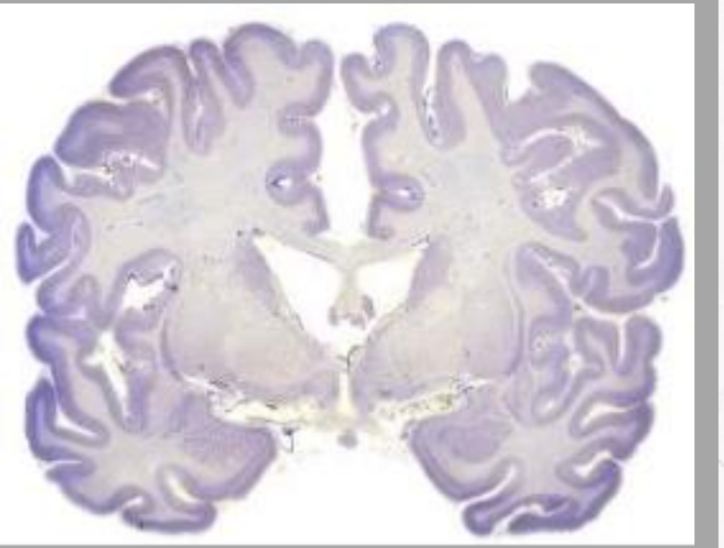
http://neurosciencelibrary.org/Specimens/primates/human/brain/human8sect6.jpg

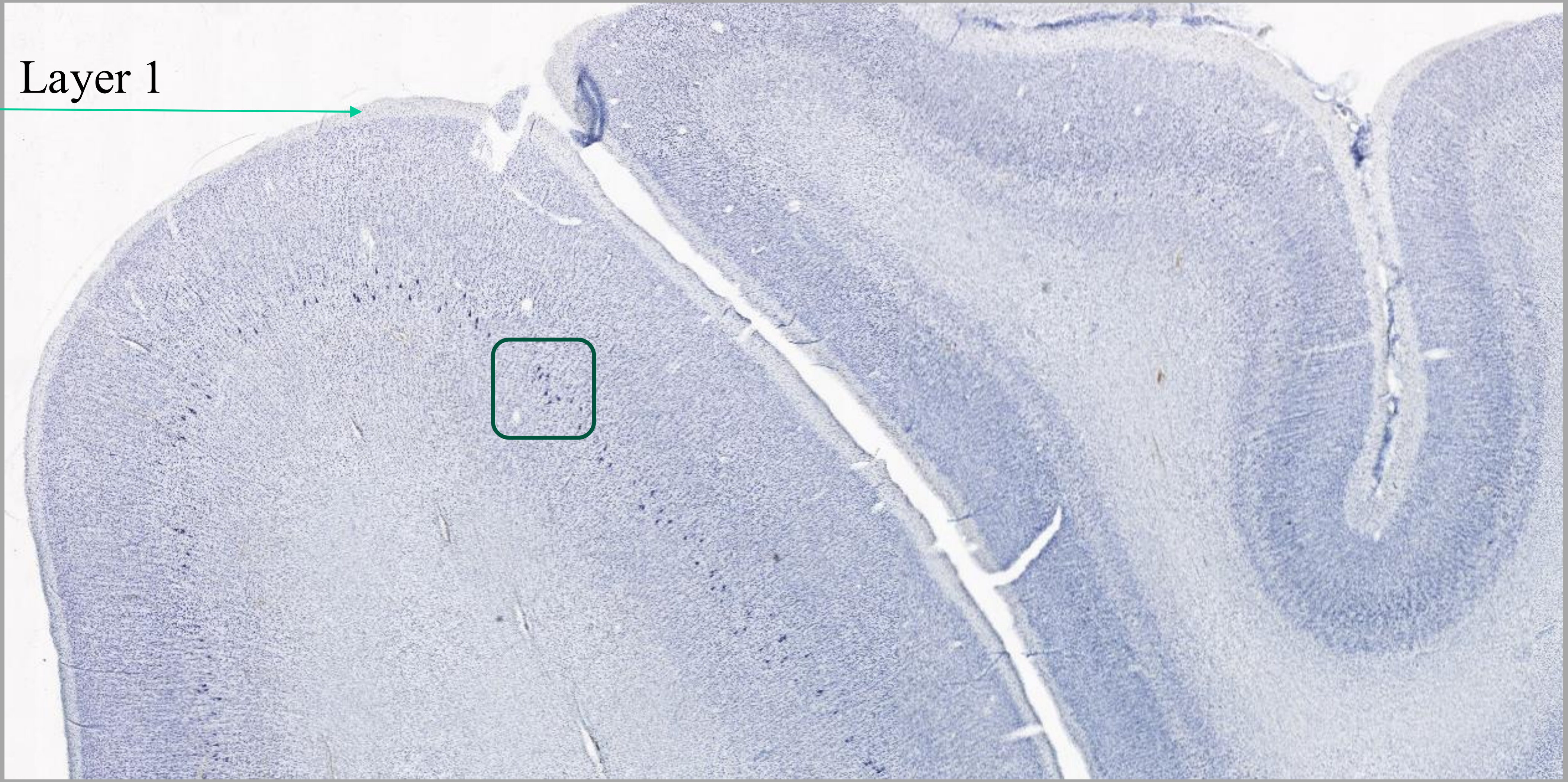
Cyto-architecture of the neocortex

- This is nissl stain

What is the distribution, density, and/or pattern of cells in the neocortex?

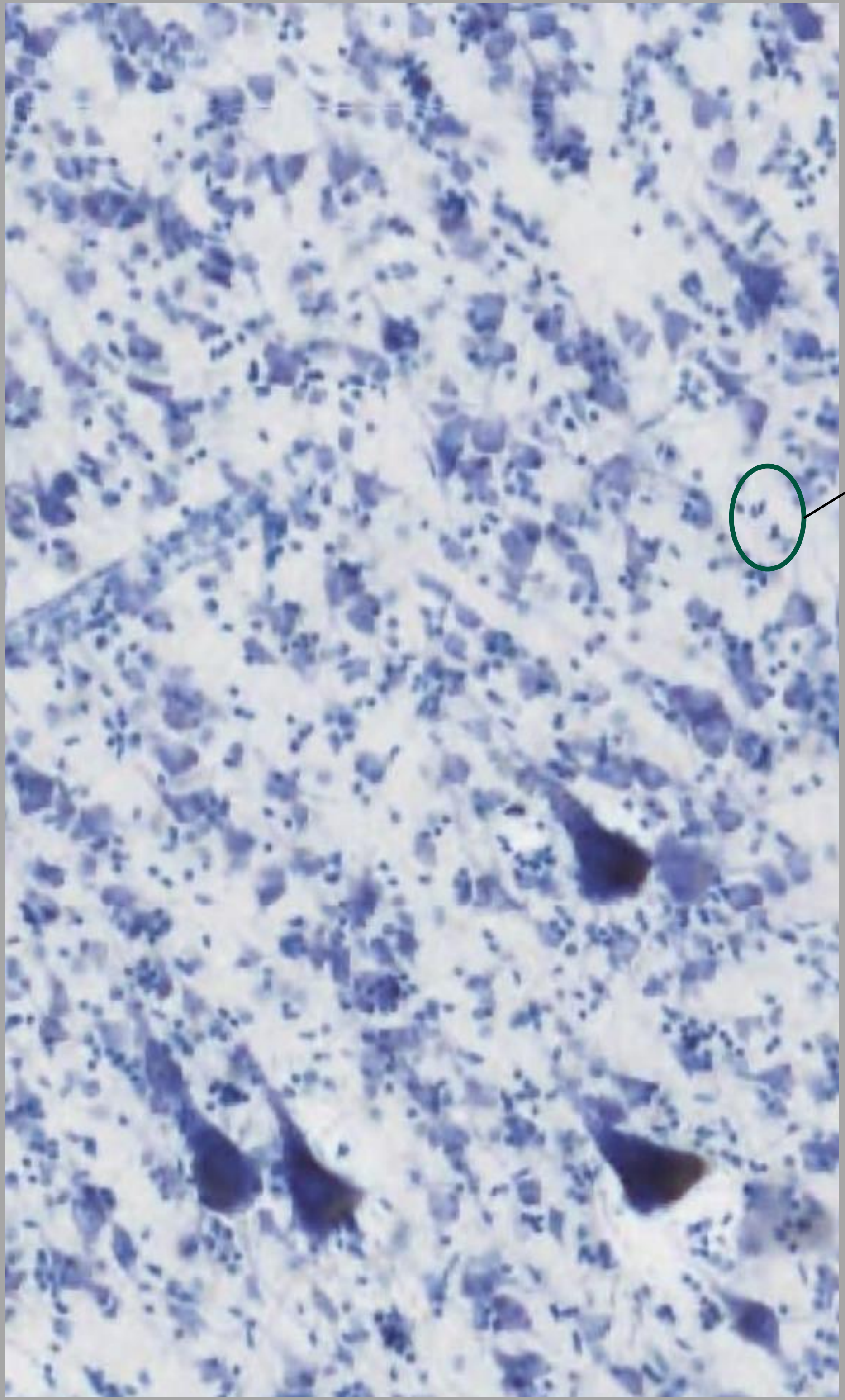
- Sulci
- Gyri





- Box shows pyramidal cells, see next slide

Neocortex has a 6-layered cytoarchitecture. Layer 1 has very few neurons and mostly axons

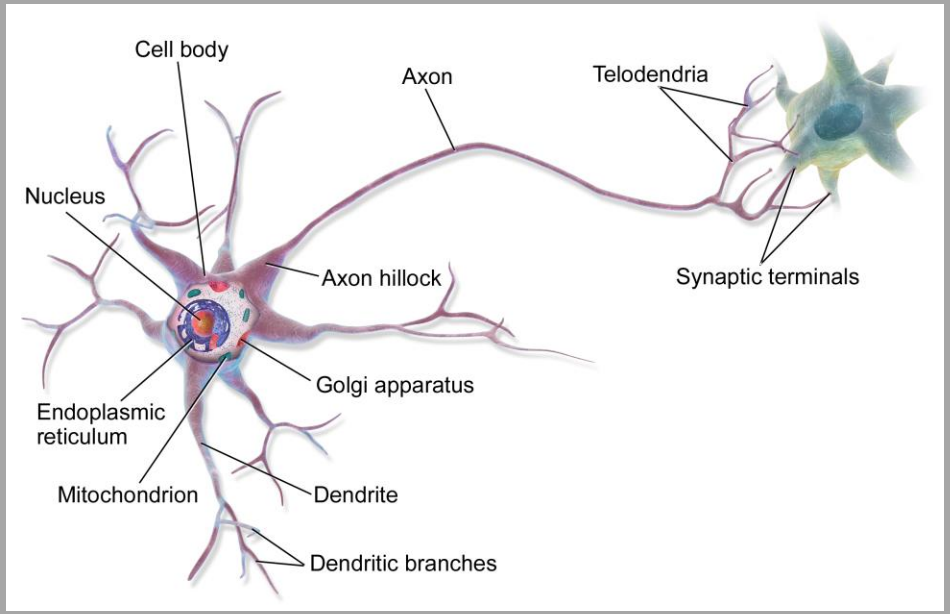


Cytology of the neocortex

~ 85% of cortical neurons are pyramidal (Make + release Glutamate, spiny dendrites)
~ 15% are non-pyramidal neurons (Make and release GABA, non-spiny dendrites) Not seen here, need different stain
~ 75% of all cells in neocortex are glia i.e astrocytes, oligodendrocyte, microglia (Circled in picture)

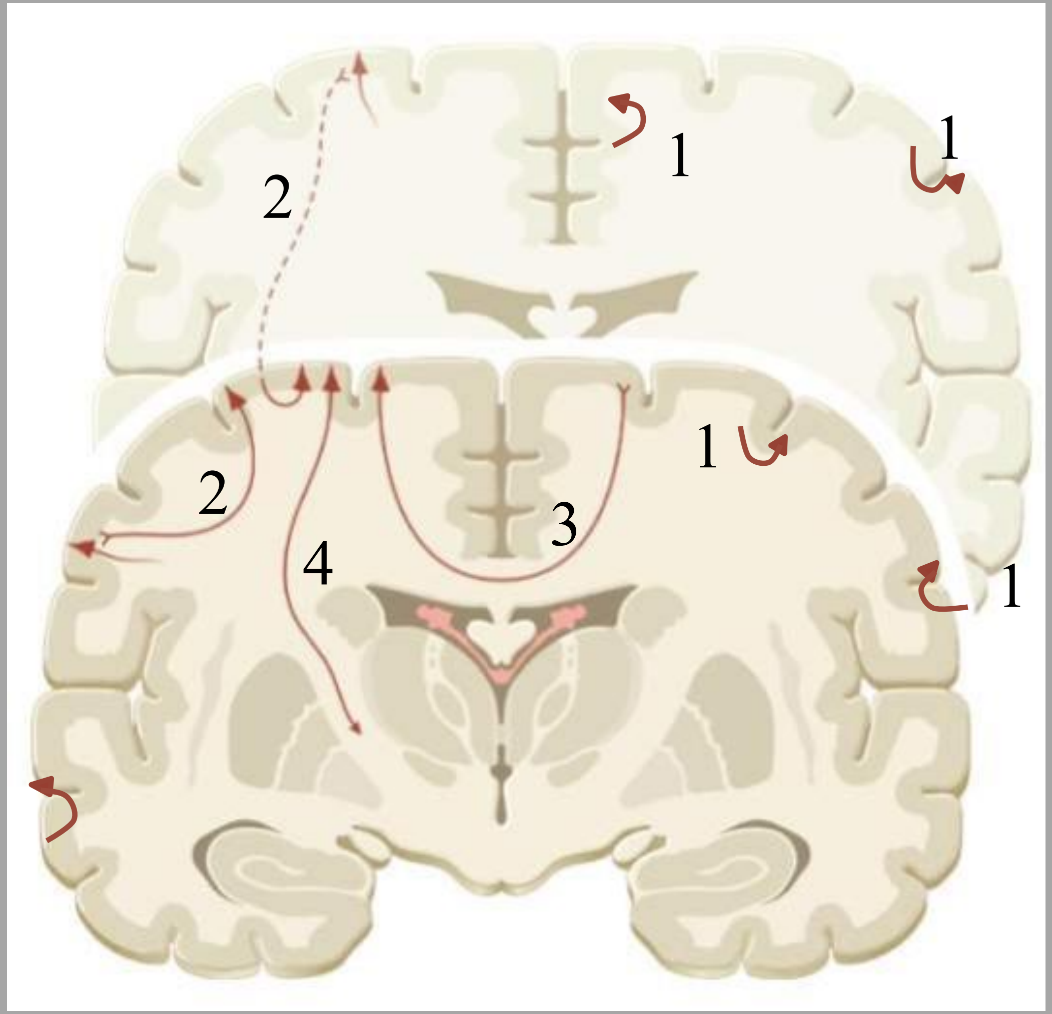
Pyramidal neuron, spiny dendrites seen

Remember parts of the neuron?



Axon going to Sub-cortical white matter

4 prototypical types of connections made by neocortical excitatory neurons:

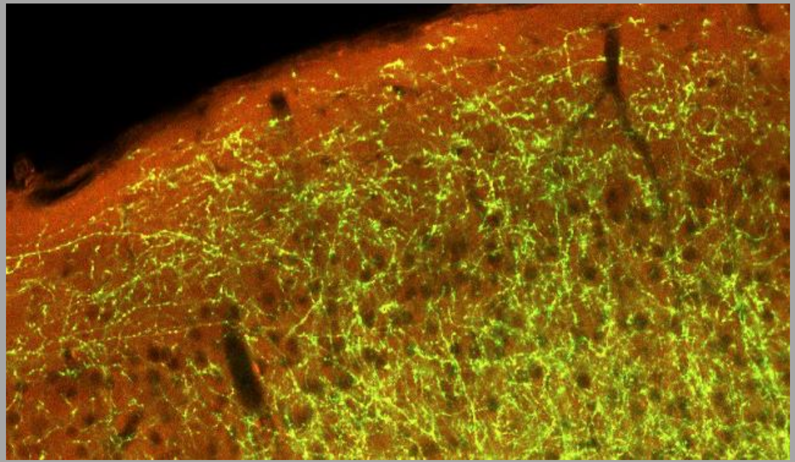


1. Local cortico-cortical
   - within one area
   - across/within layers

2. Long range cortico-cortical
- from one area to another

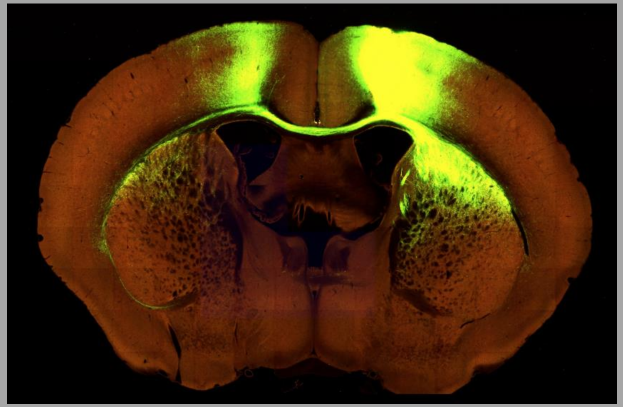
3. Interhemispheric/callosal
- generally homotopic areas in opposite hemispheres

4. Subcortical
- thalamus, tectum, basal ganglia, limbic system, spinal cord



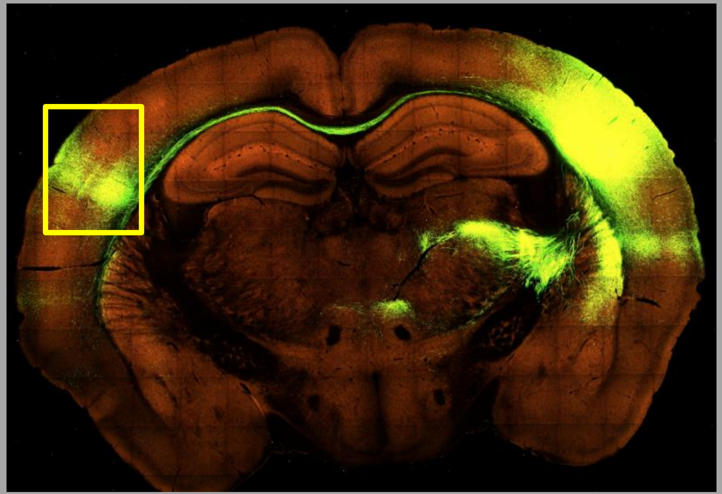
- These connections underpin the biology that drive complex human behaviors such as emotional regulation, empathy, morality, as well as sensorimotor integration etc
- Makes it very very hard to understand what neural pathways are responsible for what emotions i.e what neural pathways are aberrant in generalized anxiety disorder?

Ex 1. Motor cortex projections to contralateral motor cortex and caudate + BG

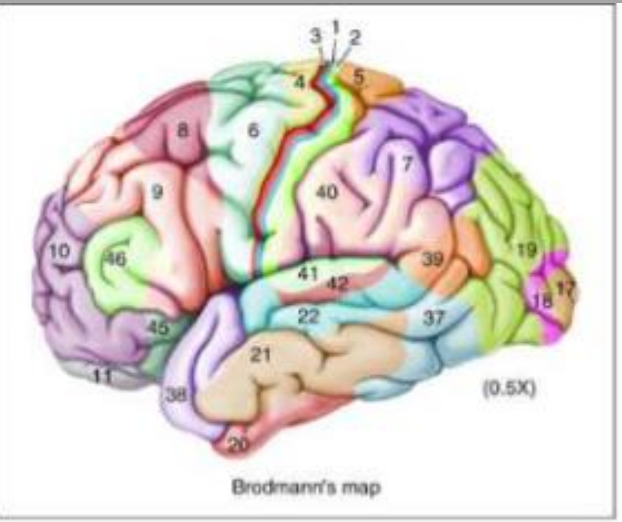


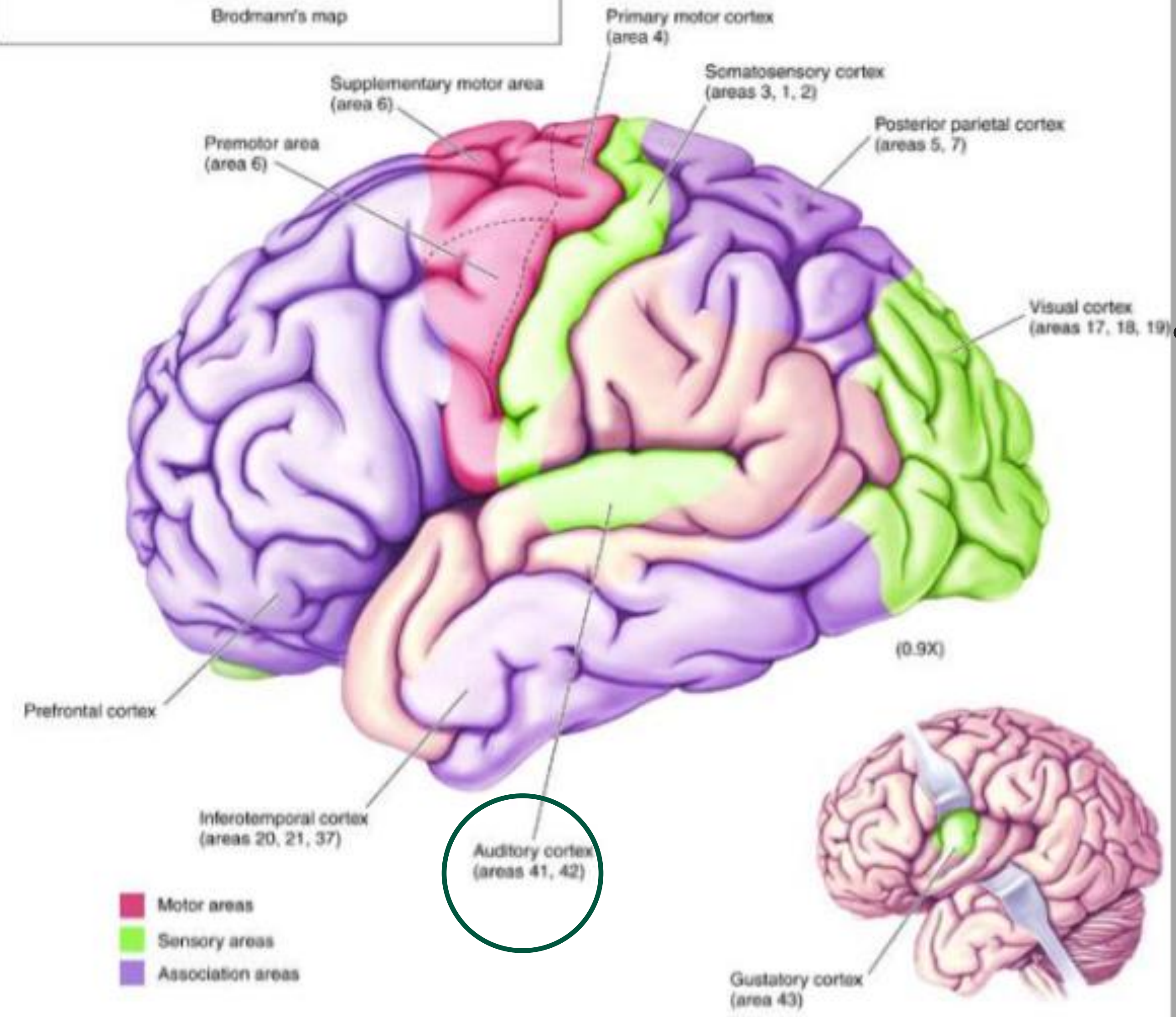
- Dye fills the neurons at the site of injection and labels the axon targets of those neurons.

Ex 2. Somatosensory cortex projections to contralateral cortex and thalamus



Understanding difference in the cytoarchitecture *** of the different lobes, gyri and sulci in various parts of the neocortex, as well as using functional maps will help one understand how the neocortex works

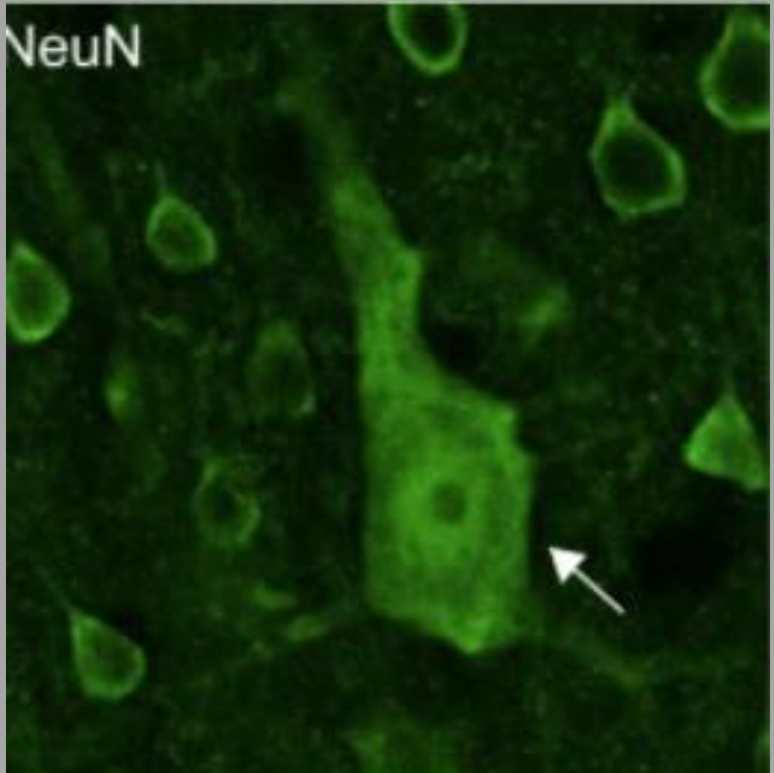




1. Major gyri are often distinct functional zones separated by major sulci.
-ex. central sulcus divides motor & somatosensory areas

2. Brodman's areas (1909) largely align with functional zones.
- ex. auditory cortex & visual cortex. About 50 areas divided based on histology of the tissue. Structure and function related, histological differences differentiate/define tissue function relatively accurately

- ***Red arrows point to Betz neurons. Unique to layer 5b of the primary motor cortex. Send projections via corona radiata to corticospinal tract. Example of unique cytoarchitecture found in only one layer of the neocortex. Affected in ALS



Functional Neuroanatomy of the Neocortex: past & present

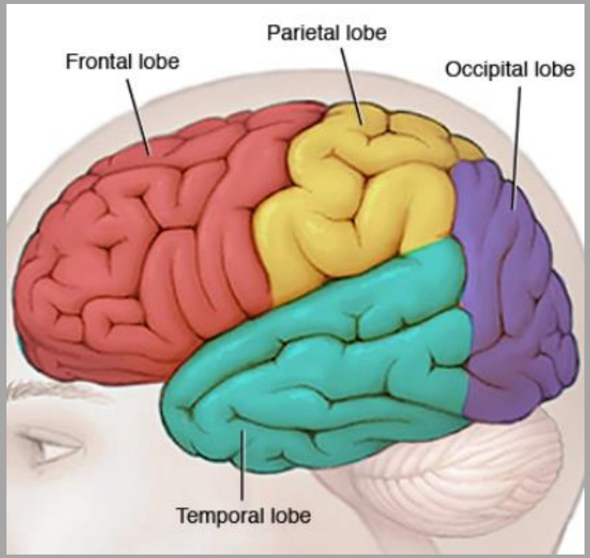
How do we learn about what the function of different parts of the human neocortex? 4 major ways

- **Clinical case reports** documented brain injury, neoplasm, vascular accident, etc. in a specific area of the brain with a correlated behavioral/clinical deficits ex) Wernicke's/Broca's area. Injury provides a clue of function

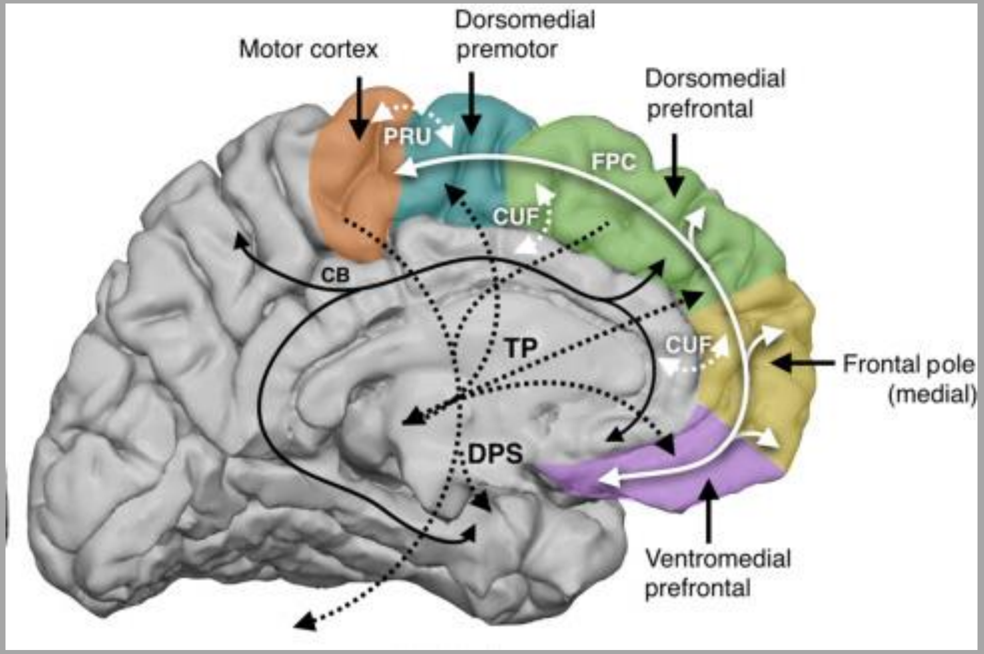
  - Requires rigorous histopathology or sophisticated imaging (MRI), not all injury or pathology is similar in extent or location among different patients (difficult to interpret)
- **Functional MRI** useful for learning about brain activity in specific regions during behavioral, sensory or motor tasks
- - Behavior limited in MRI environment and resolution is limited. Does not record single neuron, rather large groups of neurons
- **Electrical brain stimulation** coupled with behavioral observation during neurosurgery

  - Stimulation probably being performed during surgery for tumor or epilepsy treatment. Helps surgeon also to know what pathways to possible avoid in surgery, ex) don’t cut out or damage Broca's area when excising this CNS lymphoma. Limited regions sampled and not a healthy brain. Studies have caveats with this technique
- **Electrical recording of neuronal activity** during behavior/cognitive task

  - Recoding probably being performed during surgery for tumor or epilepsy treatment. Limited regions sampled and not a healthy brain. Both stimulation and recording can be done before a surgery to remove some lesion

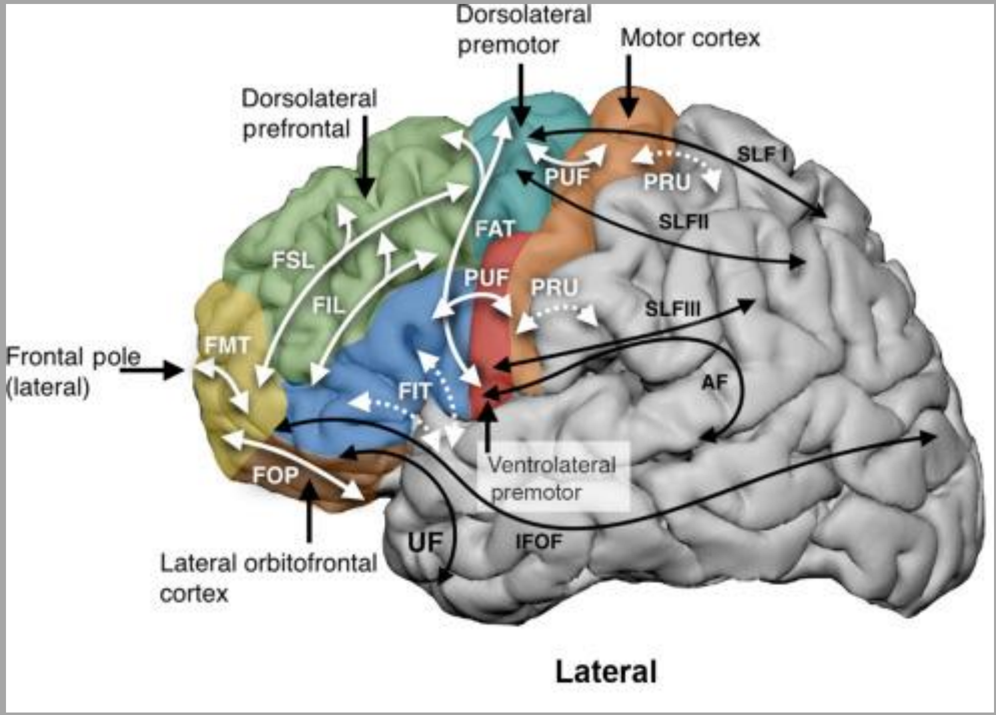


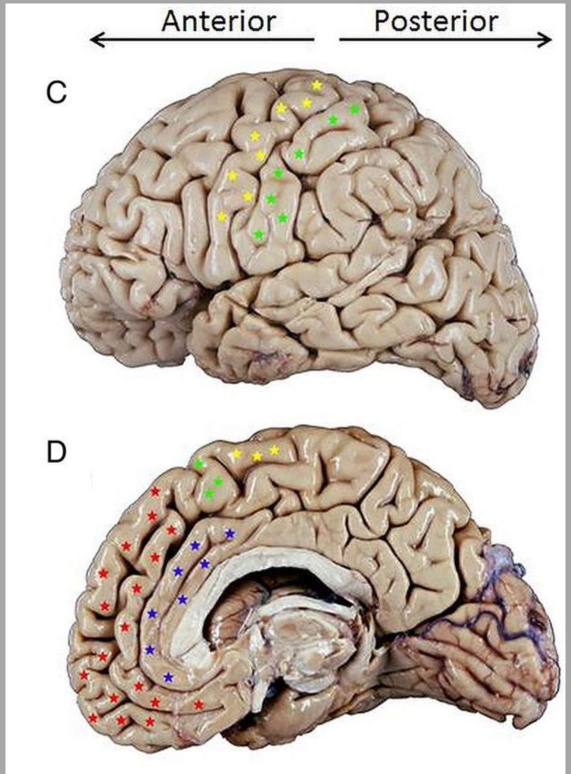
Frontal lobe/PFC: executive & voluntary motor function including speech, emotion, cognition.



All cortical areas are connected to numerous other cortical areas as well as subcortical areas

- Executive functions (EF) are a collection of distinct but related cognitive functions that typically refer to aspects of cognition and behavior required to successfully **execute complex tasks** (Denckla, M. B. 1996).
- Frontal lobe/ PFC functions include but not limited too: **setting and maintaining goals, planning, delayed gratification, organizing, initiating, inhibiting, self-monitoring and evaluating, thinking and acting strategically, problem-solving, shifting flexibly among cognitive and affective sets, deliberately controlling cognitive functions, and managing moment-to-moment events.**
- Behavior at work vs with family
- A **cognitive set** governs a person's thinking and information processing, while an **affective set** guides their emotional responses
- Also remember the pre-motor cortex and supplementary motor area are in the frontal lobe, which means frontal lobe has a role in planning, organizing and executing complex motor tasks





SC Lanata & BL Miller 2016

bvFTD; behavioral variant FrontoTemporal Dementia/Degeneration

Deficits in Executive Function
Poor decision-making, judgment, problem-solving, and organizational skills. Examples include: Difficulty planning the day's activities, Questionable financial decisions, On-the-job mistakes.

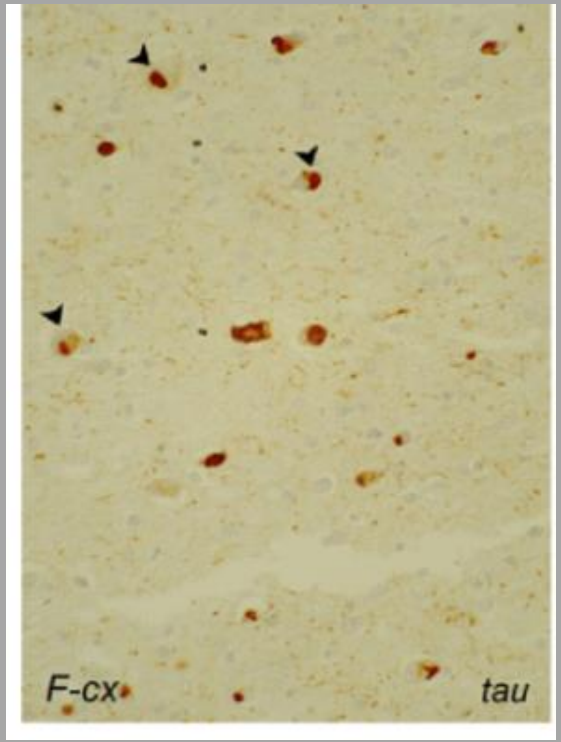
Disinhibition: A loss or lack of restraint based on social norms, leading to inappropriate behavior and impulsivity. Behaviors may include:
- Making uncharacteristic rude or offensive comments → Boards questions will usually include this in stem
- Ignoring other people's personal space, reckless spending
- Touching strangers or inappropriate sexual behavior
- Aggressive outbursts or Apathy
- Onset is usually between 45-65 years old. Younger than Alzheimer's Disease

Indifference or lack of interest in previously meaningful activities. Behaviors may include: Loss of interest in work, hobbies, and personal relationships, Neglect of personal hygiene.

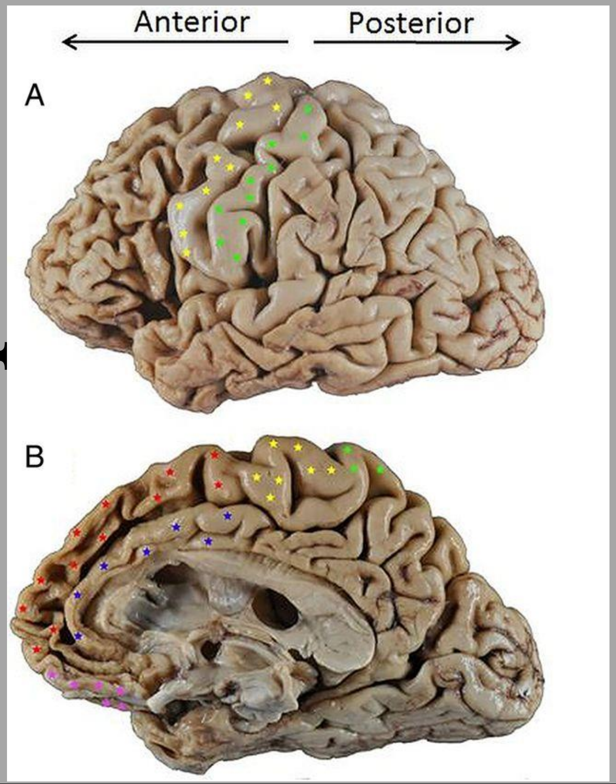
Loss of initiative, Emotional Blunting, Loss of warmth, empathy, or concern for others. Behaviors may include: Indifference to important events (e.g., death of a family member or friend); Failure to recognize that loved ones are upset or unhappy, Compulsive or Ritualistic Behaviors, Single behaviors or routines that are performed over and over.

Adapted from https://www.theaftd.org/

Tau aggregates



bvFTD patient

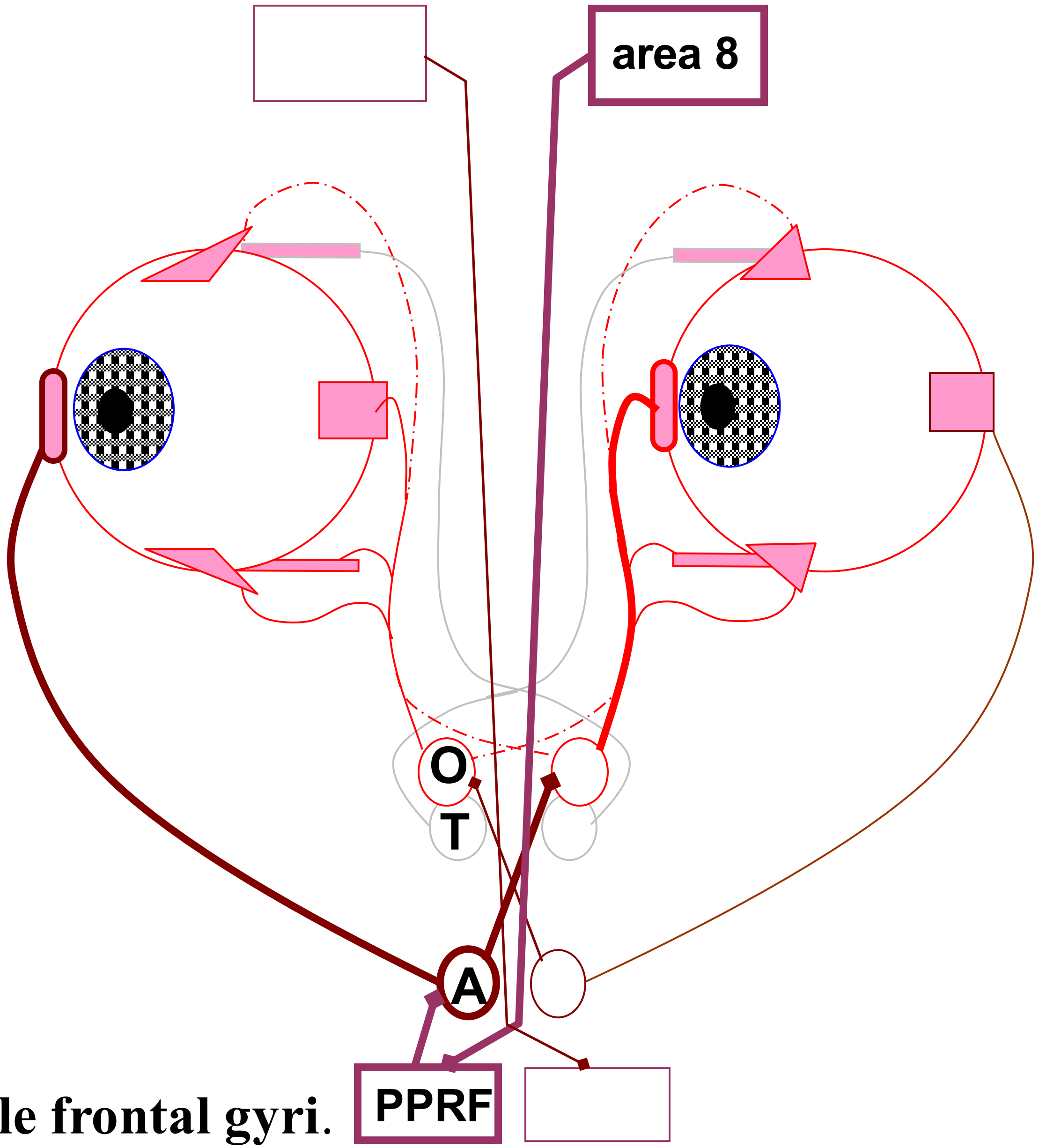


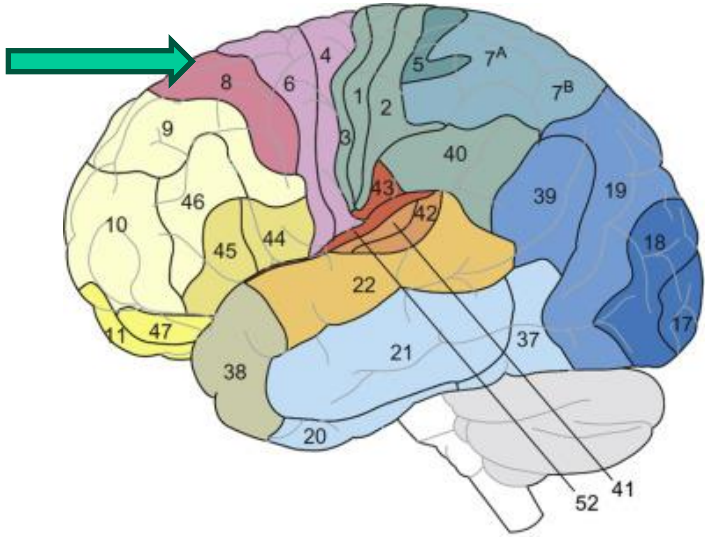
Caused by mutations in MAPT - tau

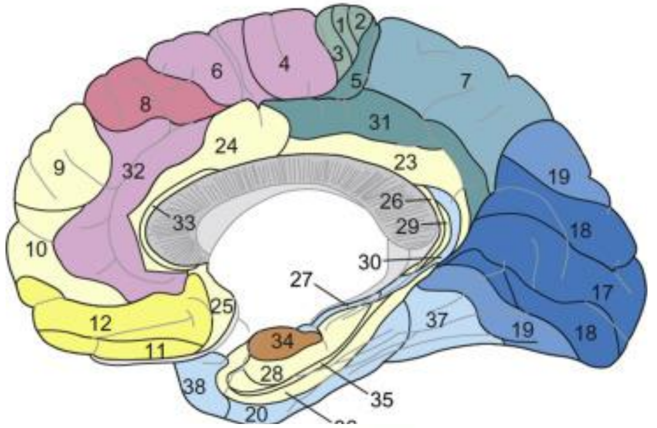
Frontal lobe area 8

- Vital for rapid, saccadic voluntary eye movements

- *Cortical **area 8**, Sends projections to brainstem (**P**aramedian **P**ontine **R**eticular **F**ormation) → PPRF then projects to abducens and oculomotor nuclei to carry out eye movement*





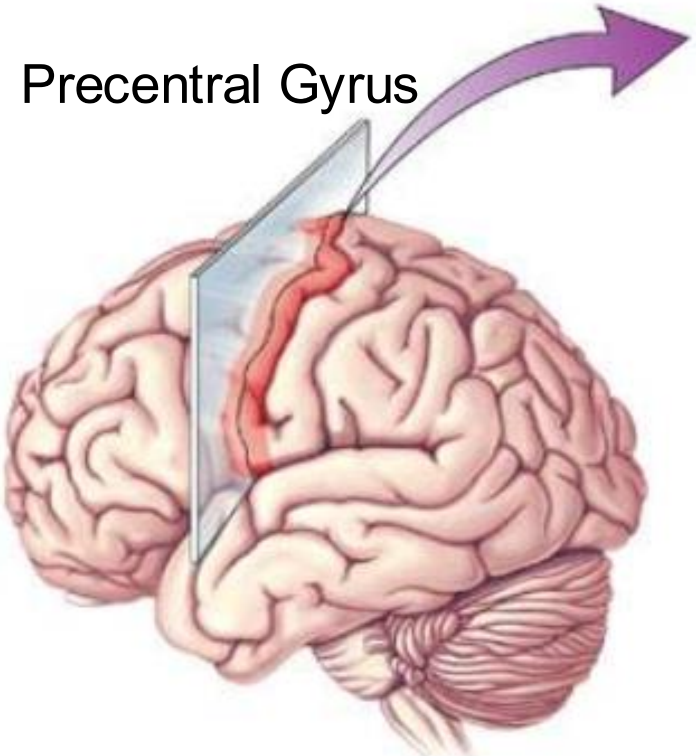


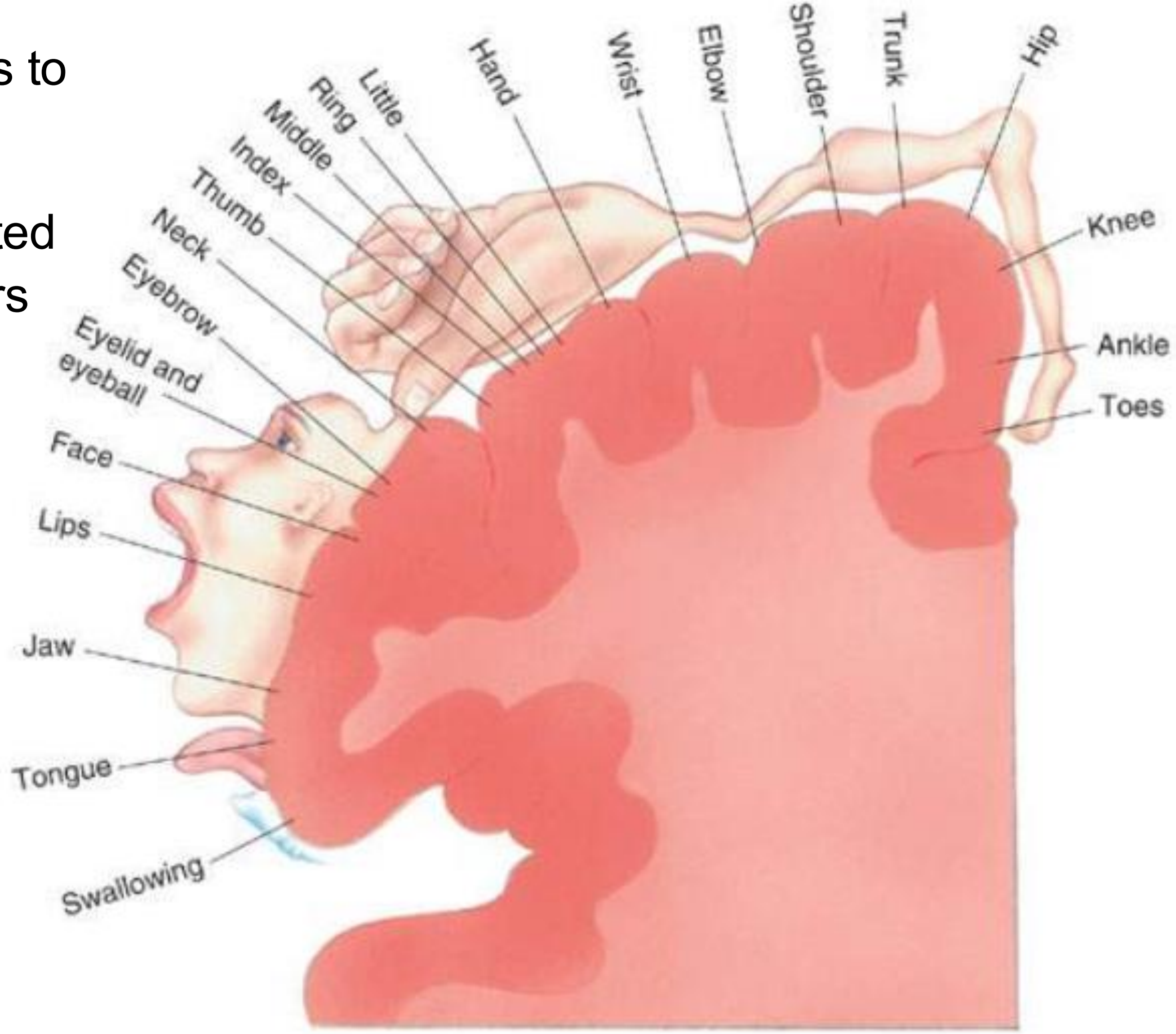
- Found in parts of **superior and middle frontal gyri**.
- Receives inputs from visual cortical areas.
- Electrical stimulation causes eye movements.
- Electrical recordings show action potentials in area 8 neurons before eye movements.
- Area involved in imitation/planning of the eye movement
- Lesion/injury to Area 8 cause deficits in voluntary eye movement

Primary Motor Cortex

- Produces voluntary motor movements via projections to spinal cord
- Many more neurons devoted for movement of the fingers and face musculature.

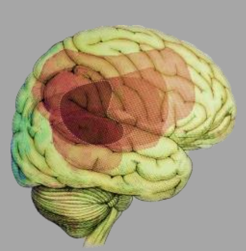
Precentral Gyrus





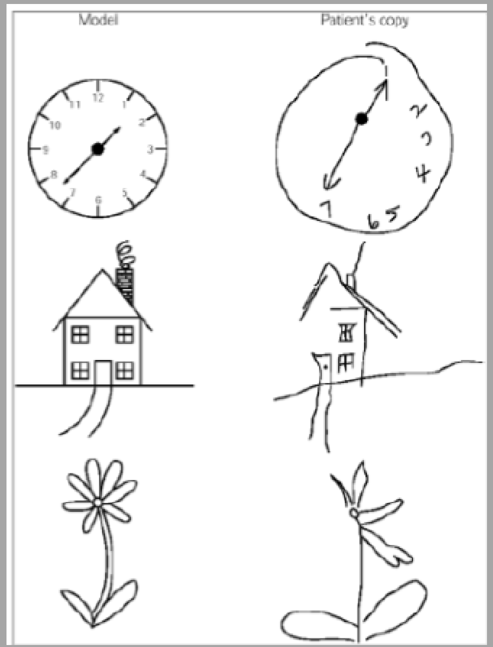
- Afferents from motor, vestibular, sensory systems. i.e catching a football with your hands or catching yourself from tripping down the stairs
- Efferents to thalamus, brainstem, spinal cord, basal ganglia for voluntary motor movements.
- Electrical stimulation causes muscle twitches.

Right Posterior parietal lobe damage: Spatial Hemineglect

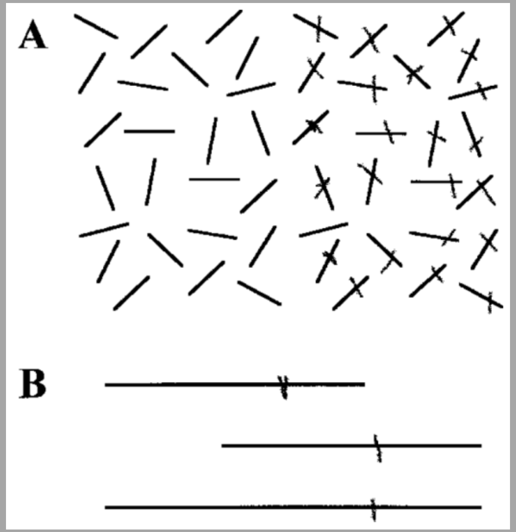


- Hemi-neglect: Lack of **attention** toward one side of the body & one side of the world. Will put clothes or makeup on one side of the body. **Not a vision problem. This is an issue of higher-level processing of the afferent incoming sensory stimuli.**
- Usually seen with large lesions/ injury to **right (usually non dominant) posterior parietal lobe**. Artery supplying this brain region is the Right MCA. Right posterior parietal lobe involved in spatial processing for both left and right sides.
- Right posterior parietal lobe seems to play role in visual appraisal/ attention to world around us. Primary visual cortex has projections to parietal lobes. (Another lecture)
- May present after a R MCA stroke

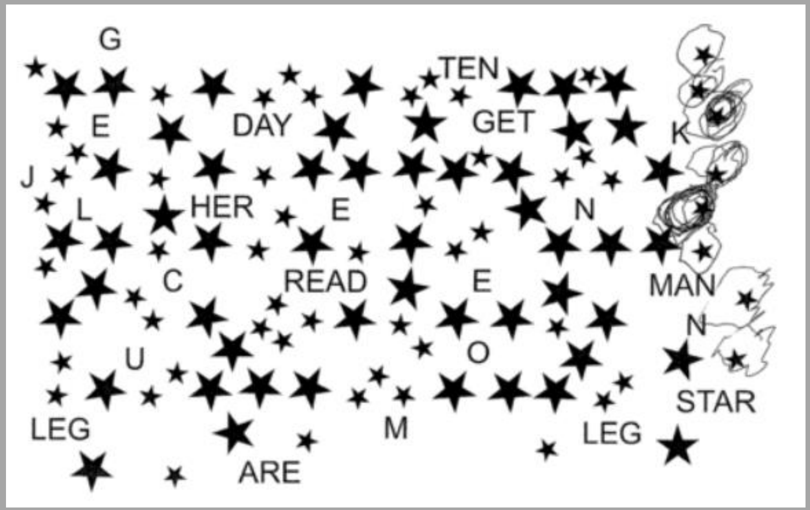
- Patient asked to copy of model picture shown left.
- Pts try to copy from memory, but they draw the clock/house/etc that is missing the left hand side of the images.



- Patient asked to draw line through middle of lines. Left hemineglect



- Patient asked to circle all small stars.

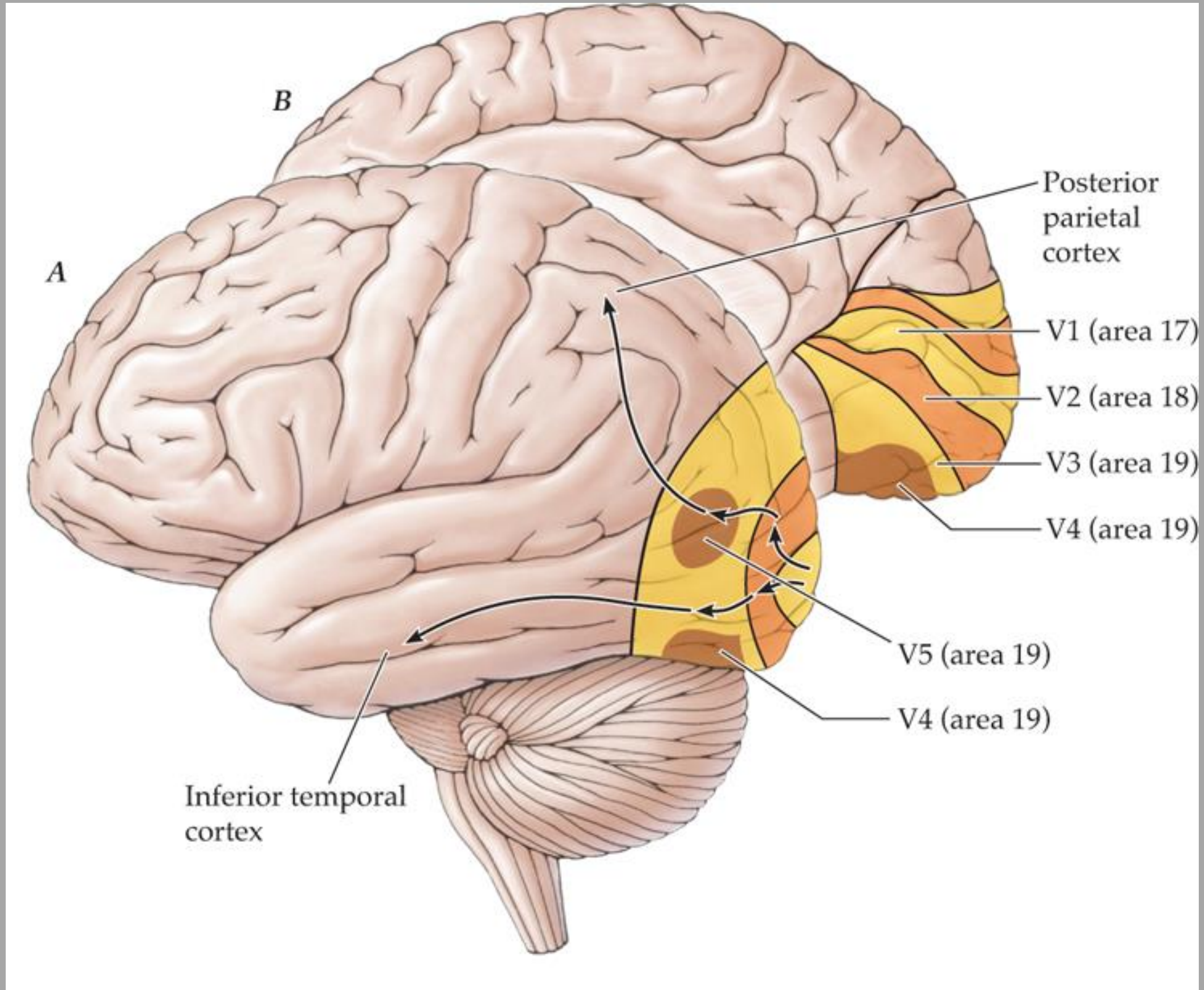


****All the tests shown are administered by putting the paper right in the middle of their face so that left and right visual fields can be assessed. When this Is done, the patient only fills out one side of the paper, likewise in the crossing the lines task or the circle the stars task.**

https://pn.bmj.com/content/15/5/333.full

Occipital lobe: visual function

- Arrows show how visual information created in visual cortex can project to areas in parietal and temporal lobes involved in higher order functions related to visual inputs.



Source: John H. Martin:
Neuroanatomy Text and Atlas, Fourth Edition,
http://neurology.mhmedical.com

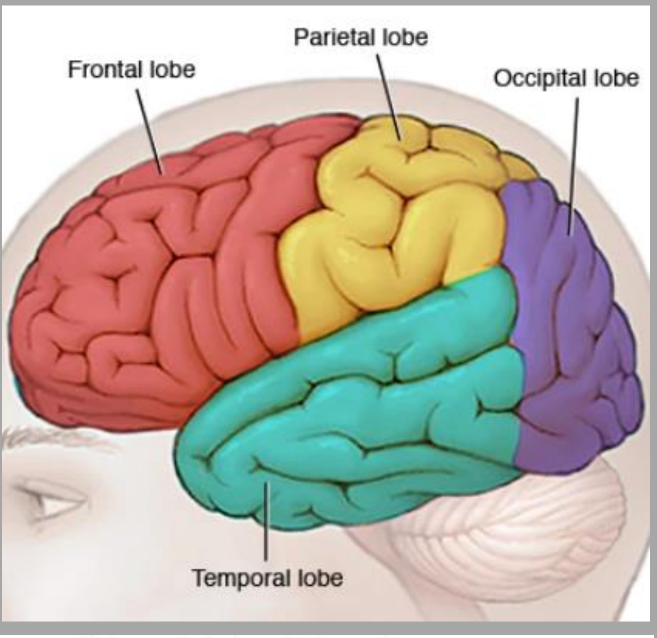
- Occipital lobes projections to higher order processing areas
- Creation of image in the occipital lobe, appraisal/processing occurring elsewhere
- Much more info in other lectures

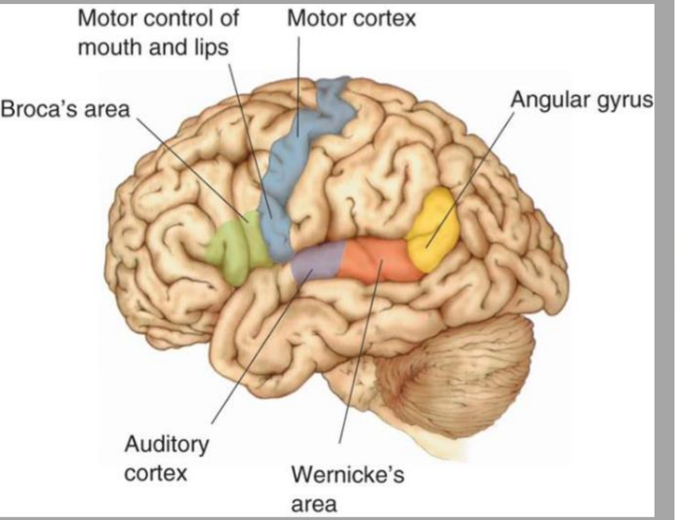
Shape
Size
Location
Orientation
Spatial frequency
Color
Movement…

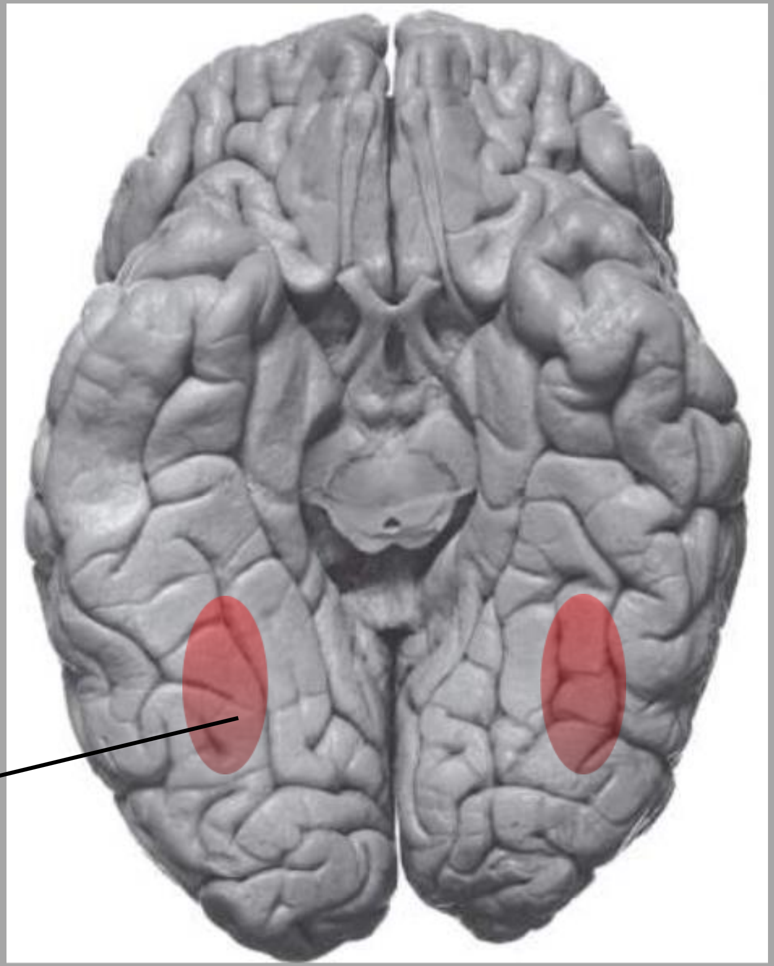
topic of future lecture…

<u>Temporal lobe</u>:

Audition (future lecture) & memory , higher order visual processing, spatial navigation.





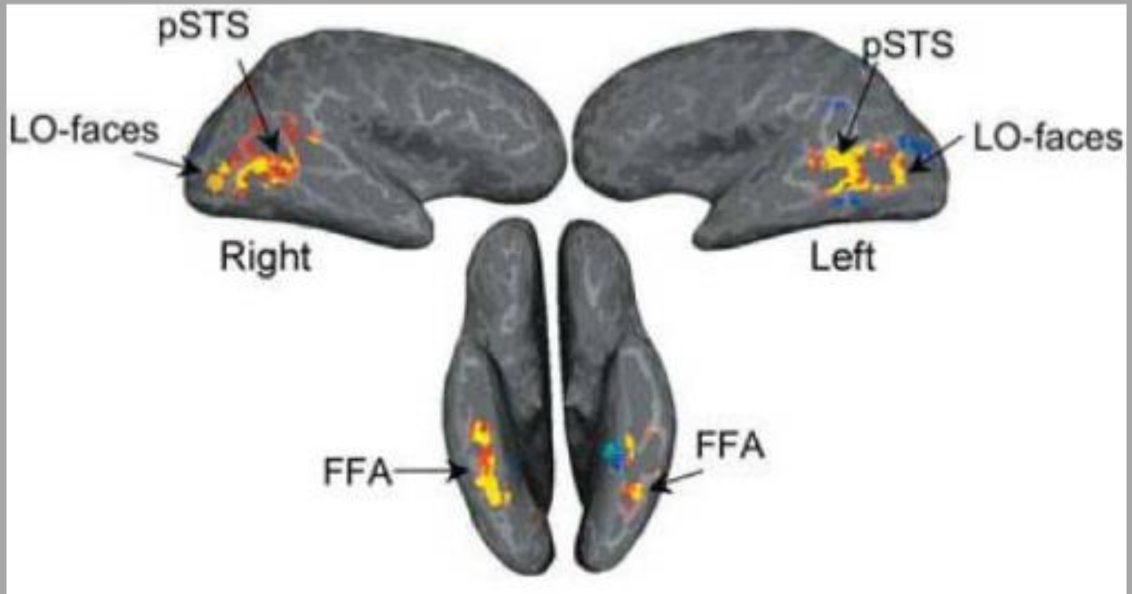


FFA damage

Prosopagnosia: Inability to recognize faces after damage to inferotemporal lobe (FFA).Blur of face when pt is asked. Still recognize voice of person. Different neural pathway involved with that

Temporal lobe areas that show increased activation following presentation of faces:

1. Fusiform Gyrus/Face Area (FFA),
2. Superior Temporal Sulcus (STS)
3. Lateral occipital area



Video

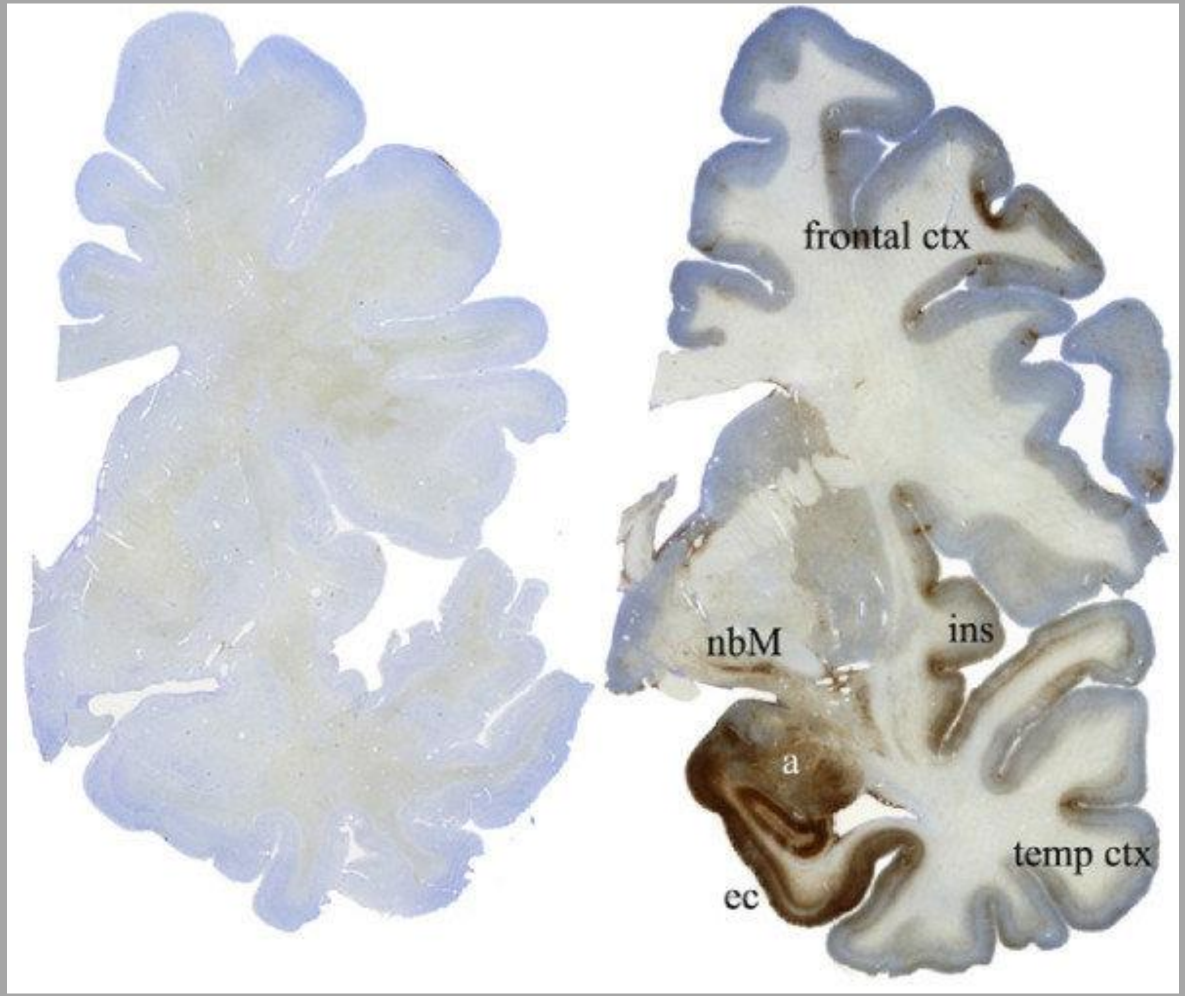
Medial Temporal Lobe

MTL vulnerable to injury after head trauma

Tau staining → neuronal injury marker

Normal

Pro football player



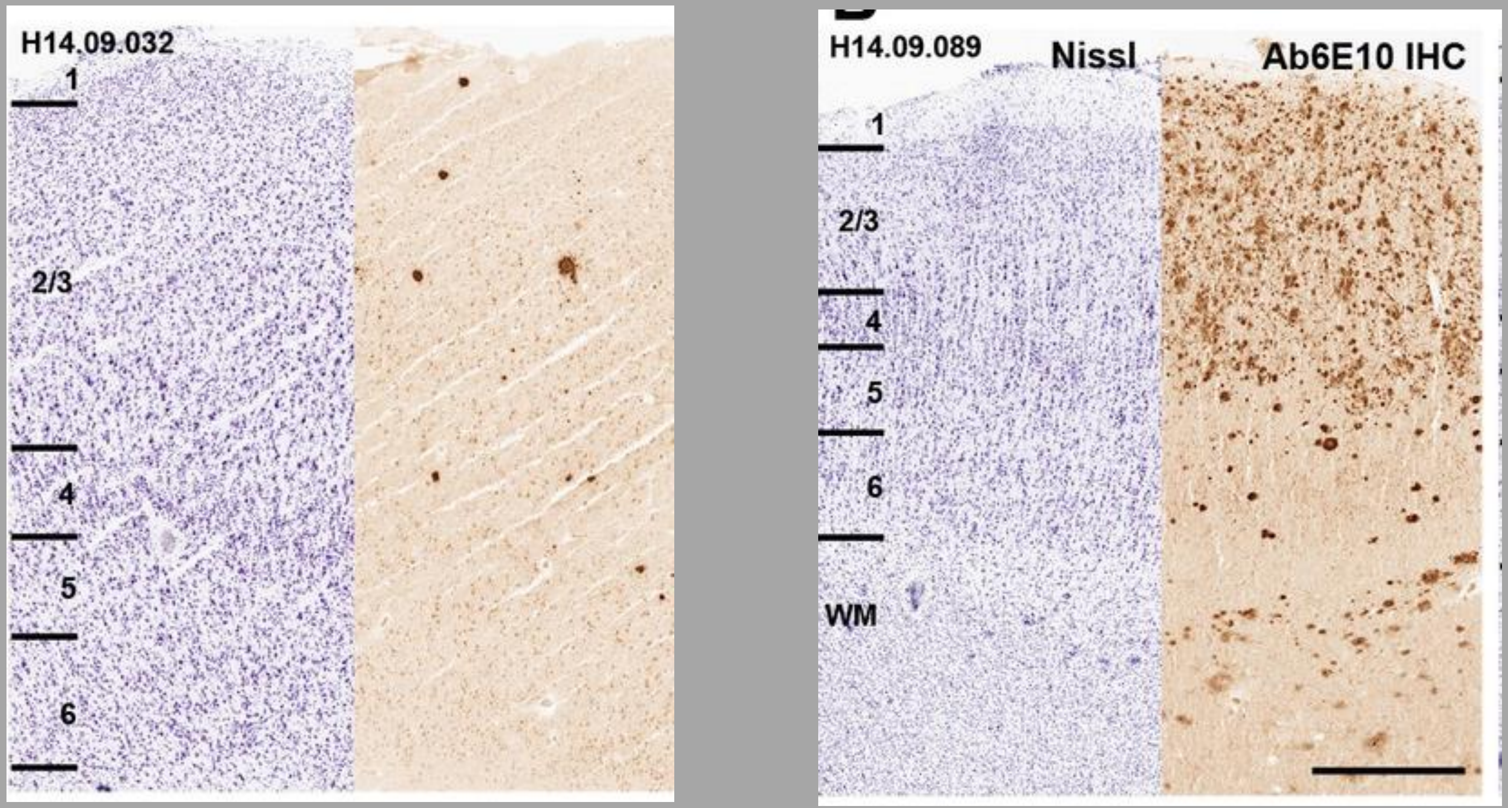
Stern et al. 2011 / 2012 Mar 12;1:2. doi: 10.1186/2193-1801-1-2.

MTL vulnerable to injury in Alzheimer's disease

- Whether Tau accumulation is the cause of memory loss or some event preceding tau creation is essentially unknown at this point
- Damage/neuronal pathology correlated with loss of memory function

Normal

Patient w/ AD



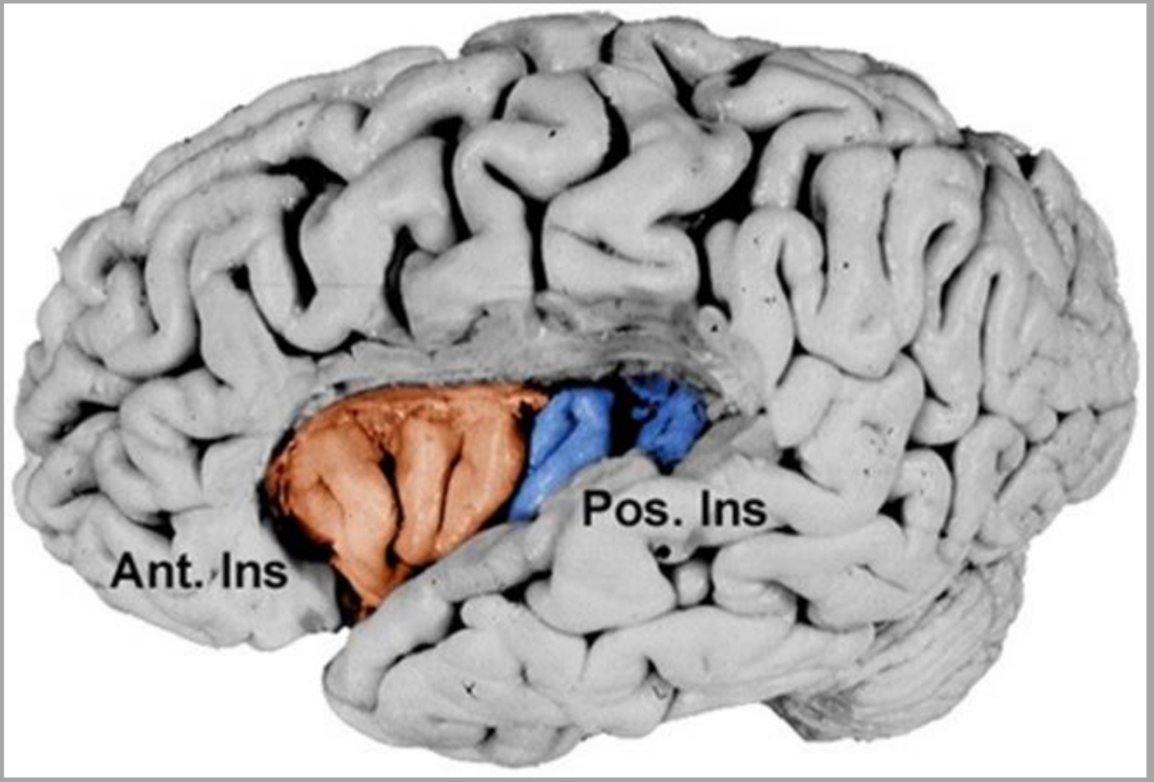
Neuropathological and transcriptomic characteristics of the aged brain. Miller JA, et al. Elife. 2017. PMID: 29120328

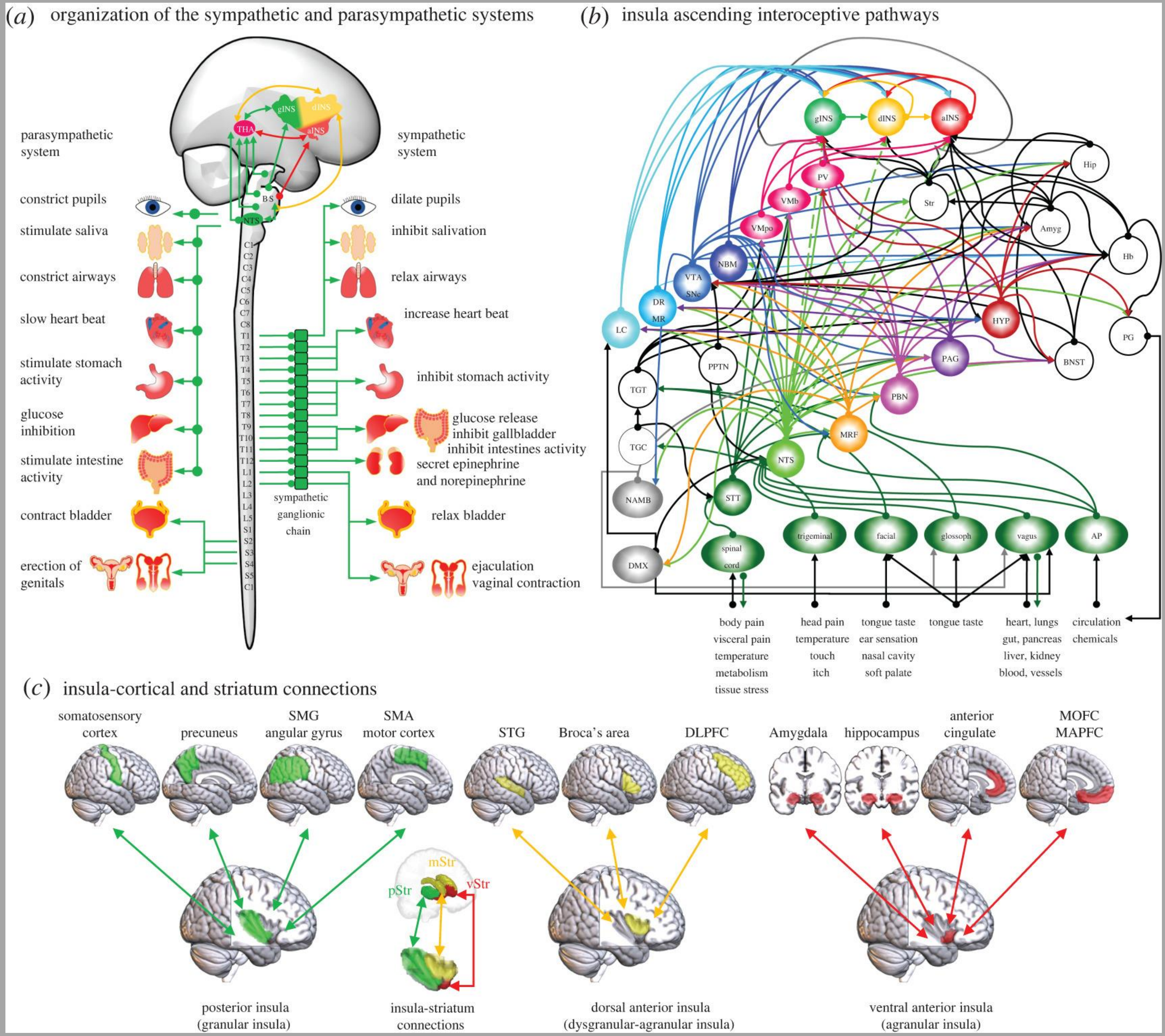
Insular cortex- diverse functions including: Gustation, olfaction, vestibular, pleasure, reward & addiction, visceral & autonomic functions. Receives input related to these areas. A hub for interoception and translation interoceptive inputs into conscious feelings. Damage to insular lobe can cause different clinical pictures.

Connectivity of the Insular cortex

Fermin AS et al. 2022

- Insular cortex is activated in tasting rancid food as well as stimuli that are deemed to be morally disgusting (violence) via fMRI. Robert M. Sapolsky. (2023). *Determined: A Science of Life without Free Will.* Penguin Publishing Group.

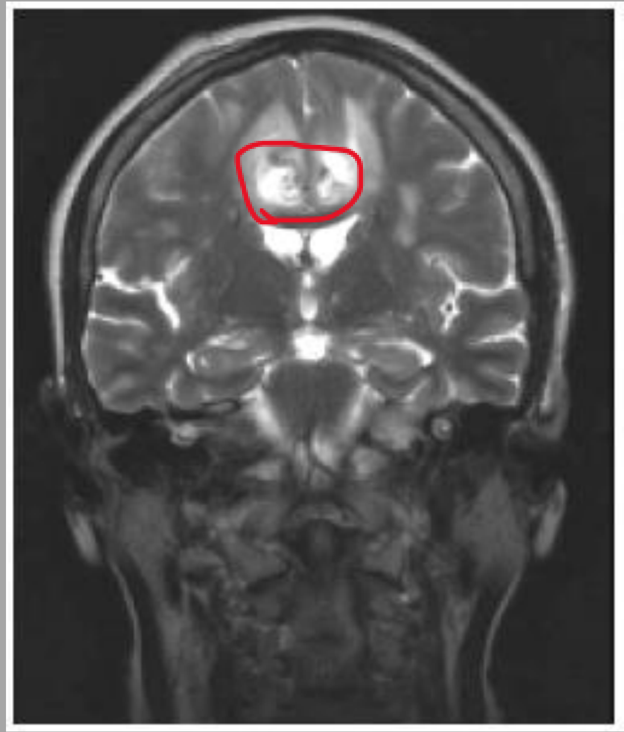




Cingulate cortex: cognition, emotion, memory (Papez circuit), 4F's

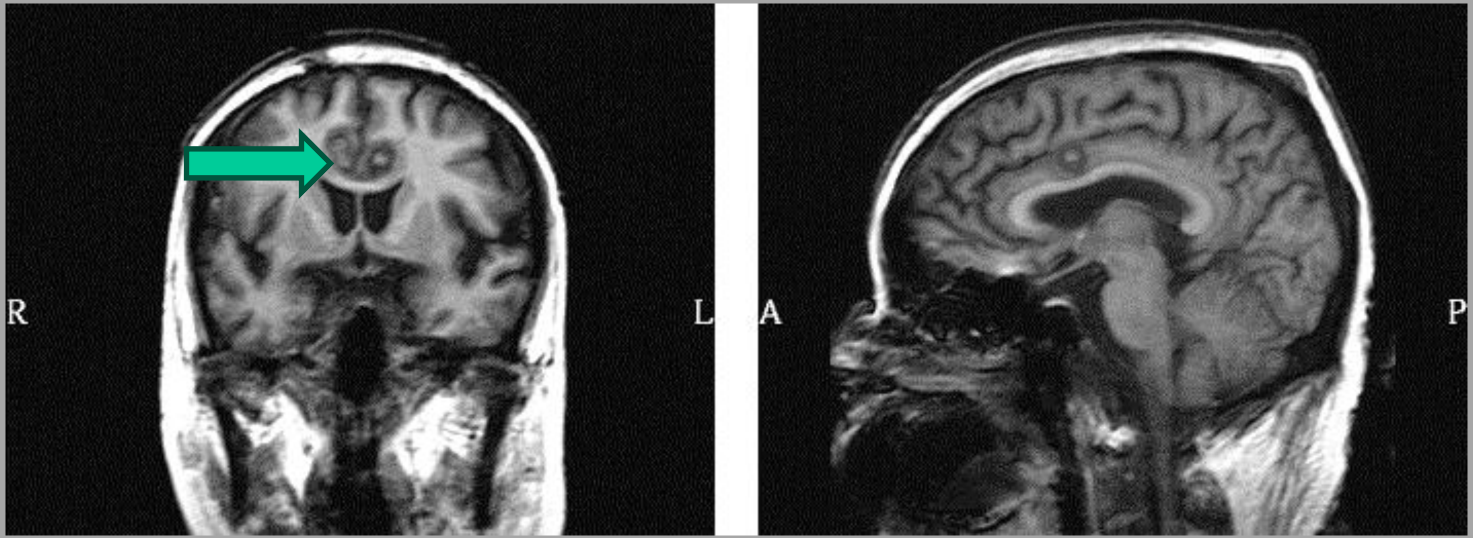
- Feeding, fighting, flighting, fornicating

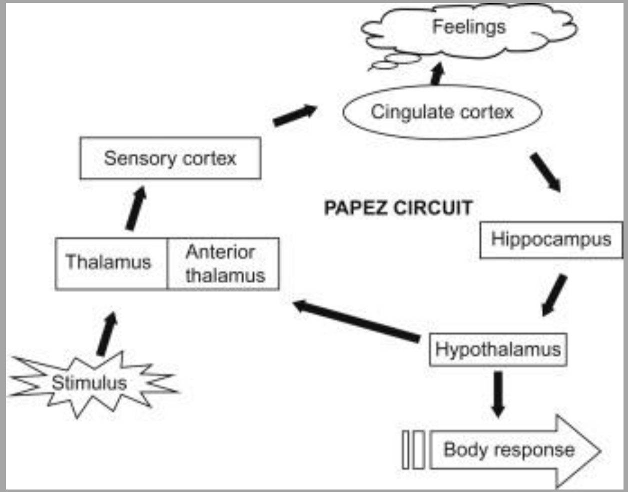
Bilateral anterior Cingulotomy for drug-resistant depression or OCD



1 wk after surgery; Sam-Eljamel 2008

Bilateral anterior Cingulotomy for the treatment of chronic pain





This patient suffered from metastatic bone pain which was completely eliminated after surgery and the effect remained for 20 months. (Yen CP et al. 2005)

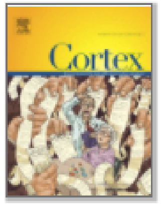
Surgery is safe and can be effective depending on numerous factors…

Cingulotomy also considered for Postherpetic Neuralgia & Trgeminal Neuralgia

ELSEVIER

Cortex

Volume 71, October 2015, Pages 323-331



Research report

Mirth and laughter elicited by electrical stimulation of the human anterior cingulate cortex

Fausto Caruana [a, b], Pietro Avanzini [b, c], Francesca Gozzo [d], Stefano Francione [d], Francesco Cardinale [d], Giacomo Rizzolatti [a, b]

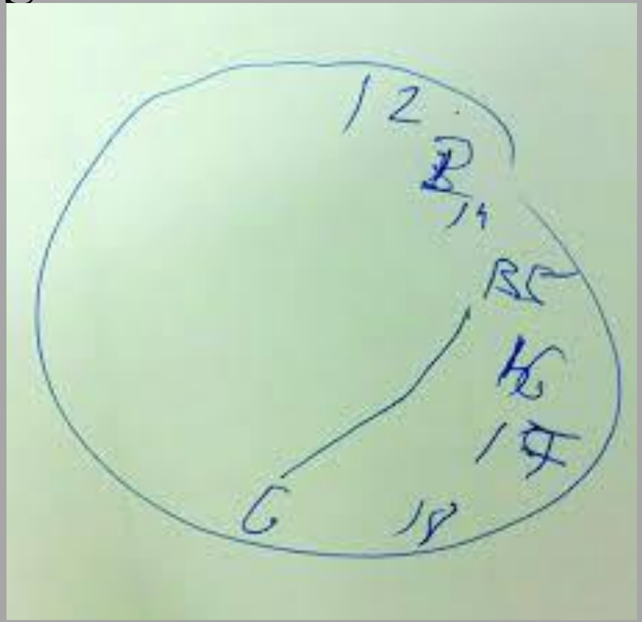
Patient	Hem.	Description
Patient 1	Right	Burst of laugh
Patient 2	Right	Burst of laugh
Patient 3	Left	Burst of laugh
Patient 4	Left	Burst of laugh
Patient 5	Left	Burst of laugh
Patient 6	Right	Laugh w/out mirth
Patient 7	Right	Laugh w/out mirth
Patient 8	Right	Laugh w/out mirth
Patient 9	Left	Laugh w/out mirth
Patient 10a	Right	Laugh w/out mirth
Patient 10b	Right	Laugh w/out mirth

Summary slide

- The entire neocortex has a characteristic cytoarchitecture based on lamina of radially-oriented pyramidal cells with diverse projections.
- Different cortical areas produce sensory, motor, or cognitive functions specific to particular modalities. These functional areas are created by the synaptic connections made within and across the neocortex as well as from synaptic input from subcortical brain regions. I.e thalamus
- Injury to a particular neocortical brain area will result in specific change in behavior or cognitive function which can be identified by a physician.

Question

- A pt suffers an ischemic stroke of the Right MCA. About a week after the stroke while the pt is recovering in the hospital, a family member noticed the pt only eating food in the right side of his plate. when asked to draw a clock he does this. What's the most likely diagnosis?



A) Prosopagnosia
B) Broca's aphasia
C) Angular gyrus lesion
D) Gerstmann's syndrome
E) Hemi-spatial neglect